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Balancing Economic Growth and Environmental Conservation: An Overview of Sustainability Practices in UAE Oil and Gas Sector

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Abstract

This study investigates the relative advantage factors influencing sustainability adoption among oil and gas service providers in the United Arab Emirates (UAE). It focuses on the relative advantage dimension to analyze six key factors: Financial Incentives, Market Share, Market Uncertainty, Return on Investment (ROI), Client Perception, and Market Competition. Additionally, the research assesses how the company type impacts its adoption of sustainability. The study utilizes a quantitative empirical approach to evaluate how relative advantage factors encourage sustainability practices among oil and gas service providers in the UAE. An online survey featuring a 7-point Likert scale questionnaire was distributed to various oil and gas companies in the country. Regression analysis was conducted to examine the influence of six relative advantage factors with respect to sustainability adoption, while the impact of company type was assessed using independent samples t-test analysis. Results indicated that approximately 60 percent of respondents believe that the proposed factors related to relative advantages, such as Financial Incentive (RAD1), Market Share (RAD2), Market Uncertainty (RAD3), ROI (RAD4), Client Perception (RAD5), and Market Competition (RAD6), contribute to the company's adoption of sustainability. Market Uncertainty and Market Competition emerged as the most significant drivers, with influence percentages between 47.5% and 48.2%. In contrast, ROI and Client Perception were found to be the least influential, with percentages ranging from 44.6% to 46.2%. Additionally, an independent sample t-test was conducted to determine if there was a significant difference between local ($M=5.07$, $SD=0.699$) and international service providers ($M=5.26$, $SD=0.612$). The t-test results ($[t(284.472) = -2.856, p = 0.005]$, 95% C.I. = $[-0.312, -0.057]$) indicated a significant difference between the two groups, assuming unequal variances. Specifically, the mean score for international service providers was higher than that of local providers, with a significant mean difference ($M \text{ difference} = -0.185$, 95% C.I.: $[-0.312 \text{ to } -0.057]$), suggesting that international companies are perceived as more substantial than local ones. The findings of this study offer valuable insights into the competitive advantages that authorities, particularly within the oil and gas sector, can leverage to encourage service providers to embrace sustainable practices. The results have the potential to initiate a transformative shift in the UAE's energy sector. This approach benefits the environment and enables a broader range of private companies and subcontractors to share the responsibility for sustainability, moving beyond the organizations that are currently involved.

Keywords: Sustainability, Oil and Gas, Relative Advantage, UAE, Sustainability Adoption, Service providers

Introduction

Nowadays, Global environmental concerns most notably climate change have grown acute, widely recognized and got high attention on all levels due to the increasing rate of anthropogenic gases emission, mainly CO₂ which imposes a negative impact on the planet, human health, and economic stability. Global warming caused by greenhouse gas emissions due to human activities has led to an increase in average global temperatures, with unfavorable effects such as rising sea levels, intensified extreme weather events, and widespread biodiversity loss. Unless significant reductions in CO₂ and other greenhouse gas emissions are achieved in the next few decades, global warming is expected to surpass 1.5°C and 2°C during the 21st century. (Intergovernmental Panel on Climate Change, 2021). Under natural conditions, sunlight reaches the Earth by passing through the atmosphere, which acts as a protective shield against harmful radiation, such as ultraviolet rays. Beyond this protective role, the atmosphere helps maintain a stable amount of heat essential for sustaining life, while allowing excess heat to escape (Boehm, et al., 2022). However, this balance has been disrupted due to human activities, particularly the extensive use of fossil fuels like oil, gas, and coal as primary energy sources. These activities have led to a significant accumulation of greenhouse gases (GHGs), which interfere with the natural release of heat into space (Shafer, 2022). Consequently, the Earth's temperature has been gradually increasing. Since 1880, the global temperature has risen at an average rate of 0.08°C per decade, accelerating to 0.18°C per decade starting in 1981. Over the past 140 years, this has amounted to a total rise of 1.52°C. According to the National Oceanic and Atmospheric Administration (NOAA), the year 2021 was approximately 0.84°C warmer than the average of the previous century, making it the sixth hottest year on record (Linsey & Dahlman, 2023). The increase in global average temperature can be seen in Figure 1 which illustrates global surface temperature anomalies from 1880 to 2023. It employs a color-coded map to show deviations from the average temperatures of the 1901–2000.

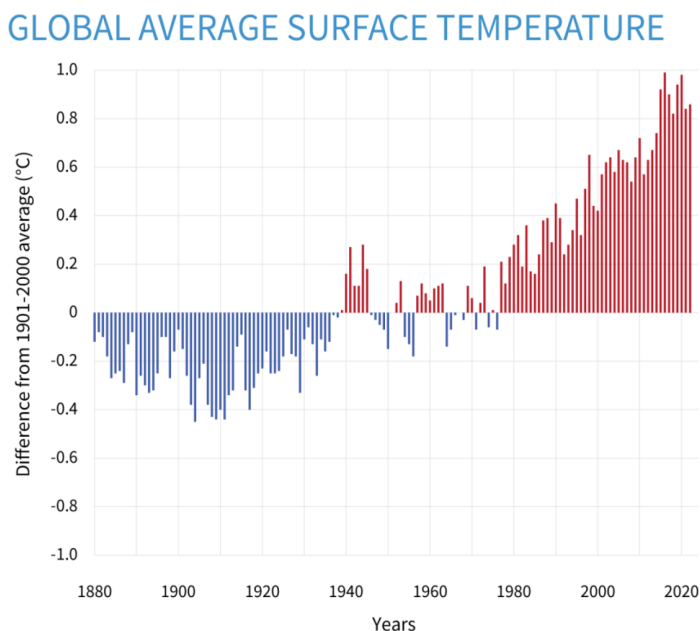


Figure 1—Representation Global Annual Surface Temperature Anomalies (1880–2023) *Note. Source: (Linsey & Dahlman, 2023, p. 1)*

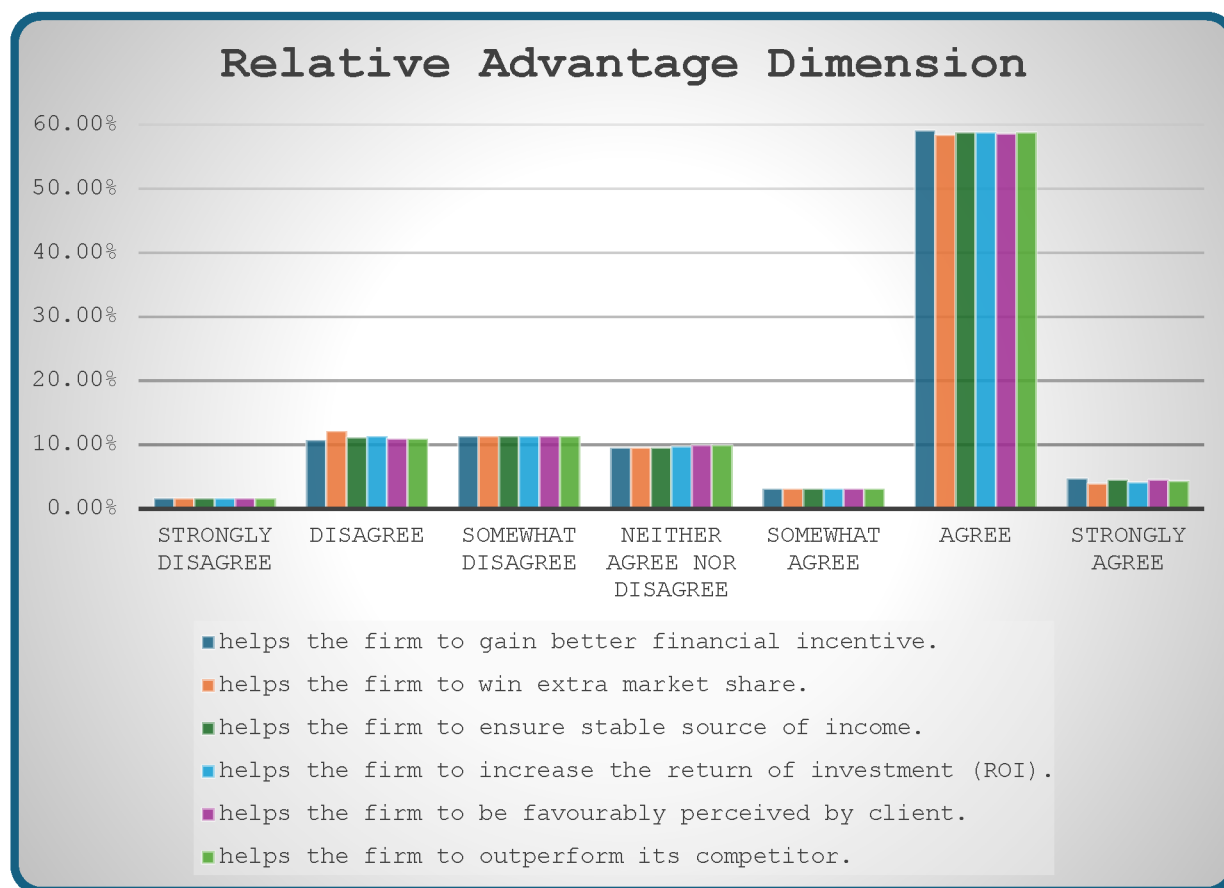


Figure 2—Response rate for Relative Advantage construct.*Note.* Source: (Author, 2024).

The Oil & Gas sector is a main contributor to environmental degradation, particularly the upstream sector activities which include exploration, drilling, and production are often associated with considerable amount of carbon emissions, water consumption, and ecological disturbances. In 2023, Extraction, processing, and transportation of Hydrocarbons accounted for roughly 15% of the total energy-related GHG emissions (about 5.1 billion tons CO₂-equivalent in 2022). After adding the percentage of GHG emissions resulted from the combustion of fuels, the Oil & Gas is linked nearly to half of all energy-related emissions (IEA, 2023). As the demands for energy keep growing and the environmental regulations tighten, upstream service providers encounter increasing pressure in order to integrate sustainability into their practices and operations. (Okeke, 2024)

The United Arab Emirates (UAE), which is widely recognized as one of the top-10 oil producers and a key player in the global energy market (Melissa Pistilli, 2024), has acknowledged the environmental implications of its hydrocarbon-based economy including greenhouse gas emissions and air pollution. Therefore, it took solid steps, and emerged as a committed regional leader in sustainability practices and climate action. As H.H. Sheikh Mohamed bin Zayed Al Nahyan remarked, "The UAE will continue its role in mobilizing international efforts to support environmental action, protect life, and preserve resources for future generations." (MBZ Quotes listing, 2023). It can't be denied that the UAE's national strategy emphasizes sustainability, as reflected in the COP28, UAE Vision 2030, the Green Agenda 2030, and the Net Zero by 2050 Strategic Initiative. These ambitious goals initiatives mark a transition toward cleaner energy, stronger environmental governance, and reduced emissions across the oil and gas sector.

While national oil companies and regulatory bodies lead this strategic shift for example, the Abu Dhabi National Oil Company (ADNOC), which has developed its 2030 sustainability strategy to address

global challenges while fostering economic growth and prosperity, upstream service providers still play a critical and vital supporting role. These companies provide the technical expertise, tools, and services essential for exploration and production, and their sustainability practices directly impact the sector's overall environmental performance. Therefore, they must proactively adopt sustainable strategies to ensure optimized investment, maintain competitiveness, in order to secure their market position.

Even with progress, there are still major obstacles in the way of completely incorporating sustainability into the oil and gas industry. In order to address the complex issues surrounding the oil and gas industry's adoption of sustainability, companies need to adopt a multifaceted, strategic approaches that incorporates stakeholder expectations, market dynamics, and financial planning. First and foremost, obtaining financial incentives is essential; in order to maintain economic viability while meeting social and environmental demands, businesses must coordinate long-term sustainable investment strategies with the assessment of risks and possibilities. At the same time, preserving or expanding market share via innovation highlights how crucial brand positioning and consumer perception are, particularly in cutthroat industries where sustainability is becoming more and more important. Businesses must be cautious yet flexible in light of the Oil & Gas market's uncertainty, timing their sustainable innovations to strike a balance between possible profits and erratic circumstances. Achieving a positive return on investment (ROI) is also crucial since it reassures stakeholders about the sustainability initiatives' profitability and justifies the initial capital expenditure. Last but not least, customer perception and market competition have a big impact on how well sustainability initiatives work. Businesses need to pay attention to what customers have to say and predict what their rivals will do to make sure their innovations live up to market expectations and maintain their relevance over the long run. These elements work together to provide a comprehensive answer to the industry's urgent need for long-term change.

The purpose of this study is to investigate the factors that encourage and hinder upstream oil and gas service providers in the United Arab Emirates from implementing sustainable practices. Financial incentives, market share, market uncertainty, return on investment (ROI), client perception, and market competition are the six main aspects that will be investigated. The study also evaluates how a company's type affects its sustainability adoption. A quantitative empirical approach is utilized for the evaluation. It is anticipated that policymakers and industry stakeholders would gain valuable insights from the findings to improve sustainability initiatives in the sector.

Statement of Theory and Definitions

Research Objectives

This research aims to assist Oil and Gas service providers in the UAE in integrating sustainability into their existing hydrocarbon business. In specific this study aims:

1. To examine the influence of six key Relative Advantage factors on the adoption of sustainability practices among the oil and gas service providers in UAE.
2. To assess the role of company type in shaping the adoption of sustainability practices.

Research Questions

The research addresses the following questions:

- What are the Relative Advantages key factors that drive sustainability adoption among oil and gas service providers in the UAE?
- In what way do company types affect the likelihood of adopting sustainability practices?
- What role does company type play in shaping the sustainability strategies of oil and gas service providers?

Financial Incentive

Similar to any emerging concept, sustainability demands proper financing strategies to penetrate and keep the development wheel running. In the old globalization model that existed before the emergence of sustainability, there was an excess of energy waste as the central feature of the model. Of course, this work identifies inefficiency, irresponsibility, and possibly damaging behavior against the environment as reasons for the call for a more sustainable model. However, the situation changed recently and that applied pressure on companies who would like to pursue sustainability and green culture. The role of finance is vital to set the plan forward whether to follow the right way of doing business by adding value for the globe and humanity or run after only beneficial returns regardless of the harmful side effects that could be generated. The role of finance can be seen in two aspects with respect to sustainability: evaluating the opportunities and recognizing the risk where both aimed to flourish the economy and aid public concerns with accelerating the transition process to low carbon era.

Traditional well-known responsibilities of finance in the linear economy such as monitoring and managing; income, expenses, cash flow, costs, profits and losses, budgeting, reviewing and predicting financial trends, analyzing markets, generating reports and others. Somehow, that meets Friedman's argument that says, "the business of business is business" (Friedman, 2007; Friedman, 1970, p.1). In his view, Friedman wanted to detach the business from its environmental and social duties and added that the government should be concerned about the other two dimensions. To some extent, Friedman's concept was acceptable earlier when economic impact was almost negligible and there was a real need to achieve progress in all industry lines. By that time, the current circumstances conflict with Friedman's proposal and the pieces of evidence prove that the world is working in unity and cannot be anything else.

As a result, finance in the sustainability domain should be able to provide a balanced plan taking into consideration the three pillars: economy, environment, and social. That happens by implementing a long-term vision focusing on the financial return and risk from the economic aspect and, from the other side, optimizing the impact on society and the environment (Schoenmaker & Schramade, 2018).

Market Share

Market share in a petroleum organization as noted in the study by Ahmad et al. (2019) refers to market share owned by the company in a domestic or international market. In the innovation Diffusion theory, market share has been considered one of the major relative advantage factors as it reflects the extent of the market which is held by the organization. Therefore, any innovation made is widely accepted if the organization which undergoes such innovation sees either an increase or no change in the market share of the organization. However, any negative fall in market share is often considered to be a negative perception towards innovation. This aspect will be adopted in the study as companies which own a larger market share have more resources and therefore can engage in operation but on the other hand, they also need to cater to the interests of the customers as brand image plays a major role in ensuring such a larger market share in a competitive industry.

Market Uncertainty

Market uncertainty is a major aspect of the petroleum industry as observed by Prabheesh and Laila (2020). Such market uncertainty can lead to a lack of adequate knowledge regarding future revenues which the firm would be able to secure in the future hence making the firm undertake a conservative path and continue with conventional aspects since innovation would require initial capital investment. Market uncertainty needs to be carefully assessed by managers as such determines the right time to adopt new innovative products or solutions for the market. Firms need to consider two aspects, namely firm performance in the market and the competitive advantage it can gain if it adopts innovative products over other competitors. Therefore, market uncertainty is considered with respect to basic market conditions and all potential benefits and advantages a firm can receive due to the launch of innovative services. However negative aspects of disadvantages

shall also be considered where even positive innovation would not lead to profitability due to negative market conditions as a result of uncertainty. Therefore, in accordance with the above discussion, market uncertainty will be considered as a relative advantage indicator and will be analyzed about its role to impact sustainability innovation with respect to oil and gas service providers in UAE.

Return on Investment (ROI)

Positive return on investment is desired by each organization and as noted by [Alhakimi and Sharaf-Addin \(2022\)](#), is the major part for organizations engaged in the petroleum sector as such defines profitability on the investment done in the project which would then be utilized to provide the return to stakeholders and engage in future projects. Oil and gas providers do face several risks both financial and operational in nature, hence it becomes important for them to have a positive and decent return on investment. ROI in mechanical terms can be suggested to analyze both costs incurred as well as the future profitability rate from the project, which becomes useful to managers as they engage in taking the decision to either implement innovative solutions or continue with the traditional approach. This aspect is broader than general profitability as it allows organizations to understand percentage gains, which becomes important as innovative solutions may require more capital investment. However, this aspect gives more focus on short-term or one financial period gain and does not consider long-term financial gains which occur in multiple financial years. Another financial aspect to consider is Return on Capital Employed (ROCE), which measures a company's efficiency by assessing profitability in relation to the capital utilized to generate that profit. However, this research focuses solely on ROI to evaluate profitability in relation to the initial investment, without considering the overall efficiency of capital usage ([Macenuddina et al., 2020](#)).

Client Perception

Any business organization does not operate for self-consumption but rather operates in order to sell products or services to customers or clients whose perception regarding different aspects of business is also important. Such perception can also influence the adoption of new innovative products or services in the organization as noted by [Keiningham et al. \(2020\)](#). Customer perception is measured by considering both 1) acceptance shown by customers towards the product and 2) the level of satisfaction which customers derive from the consumption. These two aspects are measured for both pre-innovation and post-innovation implementation in order to understand the effect of innovation. Such perception information is collected through the utilization of a feedback system and provides a customer viewpoint regarding adoption. This is important due to such clients being major revenue points for the organization and thus having the power to influence company profitability and hence decisions taken to implement or reject new innovative solutions in the long run. For the current study, client perception has been adopted as part of the observability indicator in which both organizational customers and retail customers have been considered as clients for the organization.

Market Competition

An organization should also consider the viewpoint of competitors, as each firm operates within a fixed market where several firms operate. The impact of competitors on innovation strategy development and implementation has been extensively covered in the study by [Calvano and Polo \(2021\)](#). Here, competition has been considered in both aspects, namely the response of the primary organization with respect to the adoption of innovative services by other competitors and the response of competitors to innovative services, products, or process implementation by the primary firm. The primary firm which is being considered shall consider innovation done by other competitors and the impact of such innovation on market performance. This will aid in the development of proper responses and allow managers to take decisions on whether to implement innovative solutions now or wait. But the concept of the first mover and each firm having a unique customer base despite the same industry should be considered which can influence financial performance after implementation. Since OPEC is an oligopolistic organization of nations with a cooperative price-

setting structure, competition affects it in a way that is traditionally different from organizations in a competitive environment. In this work, these dynamics shall be considered under the observability indicators of innovation diffusion theory within local and international contexts.

Description and Application of Equipment and Processes (Methodology)

Research Design

The study employed a quantitative research design to explore the Relative Advantage factors that influence sustainability adoption among oil and gas service providers in UAE. A cross-sectional survey was conducted to collect data from industry professionals, focusing on their perceptions of sustainability and the motivations behind its adoption in their organizations. The online survey garnered responses from 486 participants, providing a robust dataset for statistical analysis. The target population for this research included oil and gas service providers in UAE. Participants were selected through a simple random sampling technique to ensure non-biased representative sample. The final group consisted of decision-makers, managers, and technical staff from both local and international firms. Data was collected using an online questionnaire featuring a 7-point Likert scale to measure the relative advantage aspect. Additionally, the questionnaire included items concerning the type of company represented by the participants.

Research Hypotheses

This study investigated two primary hypotheses as part of its examination of factors influencing sustainability adoption. The first hypothesis explores the effect of *relative advantage* on the adoption of sustainability practices. To provide a comprehensive analysis, this hypothesis is divided into six sub-hypotheses, each addressing a distinct dimension of relative advantage.

The second hypothesis examines the impact of *company type*—distinguishing between local and international organizations—on the perceived feasibility of adopting sustainability initiatives. This distinction aims to assess whether organizational context influences the likelihood of adopting sustainable practices.

The following are the list of the proposed null hypotheses:

1. **H1:** Relative advantage has a statistically significant impact on the adoption of sustainability by oil and gas service providers in the UAE.
 - H1.1₀:** Financial incentive has no significant effect on the adoption of sustainability by O & G service provider companies in UAE.
 - H1.2₀:** Market share has no significant effect on the adoption of sustainability by O & G service provider companies in UAE.
 - H1.3₀:** Market uncertainty has no significant effect on the adoption of sustainability by O & G service provider companies in UAE.
 - H1.4₀:** Return on investment (ROI) has no significant effect on the adoption of sustainability by O & G service provider companies in UAE.
 - H1.5₀:** Client perception has no significant effect on the adoption of sustainability by O & G service provider companies in UAE.
 - H1.6₀:** Market competition has no significant effect on the adoption of sustainability by O & G service provider companies in UAE.
2. **H2:** Company type (Local vs International) has a statistically significant impact on the adoption of sustainability practices by oil and gas service providers in the UAE.

Presentation of the Achieved Data and Results (Results & Discussion)

Descriptive Analysis

[Table 1](#) below presents six demographic variables captured in the survey. The data indicates that the largest group of respondents (37.2%) falls within the 31–40 age range. Male participants made up the majority, accounting for nearly 71% of the total. Employees holding a BSc degree were the most frequent respondents. Notably, over 35% of participants had more than 30 years of experience, suggesting a strong base of industry knowledge and long-term exposure. Technical staff represented over 55% of the responses, while only 20% came from management-level employees. Lastly, international companies showed the highest level of participation, contributing more than 65% of the responses.

Table 1—Proposed Factors of IDT.

NO	Independent Variable	Dimension	Effect Type
RAD1	Financial Incentive	Relative Advantage	Enabler
RAD2	Market Share	Relative Advantage	Enabler
RAD3	Market Uncertainty	Relative Advantage	Enabler
RAD4	Return of Investment (ROI)	Relative Advantage	Enabler
RAD5	Client perception	Relative Advantage	Enabler
RAD6	Market Competition	Relative Advantage	Enabler

Note. Source: (Author, 2024)

The survey featured six targeted questions to evaluate various aspects of relative advantage in sustainability adoption as shown in [Table 2](#). These questions sought to understand respondents' perceptions of the benefits and improvements sustainability could bring to their organizations. The response rates for all questions were consistently high, demonstrating strong participant engagement and suggesting that the concept of relative advantage was well understood and seen as relevant by the respondents. This solid response foundation allows for a reliable analysis of how each aspect of relative advantage may impact sustainability adoption choices.

Table 2—Research Design: Independent Variables.

Independent variables	Measurement Target	Instrument
Relative advantage	Examining whether adopting sustainability is perceived as a more favorable alternative to traditional business practices, thereby acting as a driving factor for oil and gas service provider companies to embrace sustainability.	Questionnaire (7-points Likert)
Type of the company	Find the difference (if exist) between national and international companies with respect to sustainability adoption.	Questionnaire (7-points Likert)

Note. Source: (Author, 2023).

Reliability and validity of the study

Table 3 presents the factor loadings, Composite Reliability (CR), Average Variance Extracted (AVE), and Cronbach's Alpha (α). Factor loadings were considered acceptable if they were 0.5 or higher. Both CR and AVE values were evaluated against a minimum threshold of 0.7, reflecting adequate construct reliability. As shown, all values exceed 0.9, indicating excellent internal consistency, convergent validity, and overall measurement reliability.

Table 3—Descriptive analysis for demographic variables.

Demographic Variable	Category	Frequency out of 486	Percent %
Age	31-40 Years	181	37.2
Gender	Male	346	71.2
Education	BSc. Degree	346	71.2
Experience	>30 Years	171	35.2
Role	Technical	269	55.3
Type	International	325	66.9

Note. Response rate for the demographic variables. Source: (Author, 2024).

Hypothesis Testing

To analyze the data and test the research hypotheses, two inferential statistical methods were employed. First, linear regression analysis was used to examine the relationship between the dimensions of relative advantage and the adoption of sustainability practices. This method allowed for the assessment of how strongly each aspect of relative advantage predicts the likelihood of adoption. Second, an independent

samples t-test was conducted to determine whether there are statistically significant differences in sustainability adoption based on company type (Local vs. International). This test helped assess whether organizational context plays a meaningful role in influencing adoption feasibility.

Linear Regression Analysis. Table 4 presents the linear regression results for the null hypotheses H1.1 through H1.6. The analysis reveals a significant relationship between each independent variable and the dependent variable—sustainability adoption. The values for R, R², unstandardized coefficients (B), and standardized coefficients (Beta) are relatively consistent across all models, indicating stable and meaningful contributions from each predictor.

Table 4—Relative Advantage Construct Response Rate.

	Strongly Disagree	Disagree	Somewhat Disagree	Neither agree nor disagree	Somewhat Agree	Agree	Strongly Agree
	Row N %	Row N %	Row N %	Row N %	Row N %	Row N % %	Row N %
Help the firm to gain better financial incentive.	1.6%	10.7%	11.3%	9.5%	3.1%	59.1%	4.7%
Help the firm win extra market share.	1.6%	12.1%	11.3%	9.5%	3.1%	58.4%	3.9%
Help the firm to ensure a stable source of income.	1.6%	11.1%	11.3%	9.5%	3.1%	58.8%	4.5%
Help the firm to increase the return of investment (ROI).	1.6%	11.3%	11.3%	9.7%	3.1%	58.8%	4.1%
Help the firm to be favorably perceived by client.	1.6%	10.9%	11.3%	9.9%	3.1%	58.6%	4.5%
Help the firm to outperform its competitor.	1.6%	10.9%	11.3%	9.9%	3.1%	58.8%	4.3%

Note. Response rate for Relative Advantage construct. Source: (Author, 2024).

Independent Sample t-test Analysis.

Company Type

As previously discussed, the oil and gas (O&G) service providers in the study are categorized into two types—international and local—which may play a potential role in influencing sustainability adoption. To examine whether there is a statistically significant difference in sustainability adoption between these two groups, an independent samples t-test is conducted in this section. This test aims to determine whether the mean adoption levels differ significantly between international and local companies within the collected sample. H20: Company type does not imply significant mean difference in the adoption of sustainability by O & G service provider companies in UAE.

H2₀: Company type does not imply significant mean difference in the adoption of sustainability by O & G service provider companies in UAE.

An independent samples t-test was conducted to determine whether a significant difference exists between local ($M = 5.07$, $SD = 0.699$) and international service providers ($M = 5.26$, $SD = 0.612$). To assess the assumption of homogeneity of variances, Levene's test was also performed. The result ($F = 4.297$, $p = 0.039$) indicated a significant difference between the variances of the two groups ($p < 0.05$), suggesting a violation of the assumption of equal variances for the company type factor, as shown in Table 5.

Table 5—Factor Loading.

Measurement Factors	Measurement Item	Statements	Factor Loadings	CR	AVE	α
Relative Advantage	RAD1	Help the firm to gain better financial incentive.	0.991	0.986	0.908	0.994
	RAD2	Help the firm win extra market share.	0.967			
	RAD3	Help the firm to ensure a stable source of income.	0.988			
	RAD4	Help the firm to increase the return of investment (ROI).	0.977			
	RAD5	Help the firm to be favorably perceived by client.	0.99			
	RAD6	Help the firm to outperform its competitor.	0.987			

Note. Reliability and validity of the Relative Advantage construct. Source: (Author, 2024).

Table 6—Linear Regression Analysis.

Hypothesis	Model Summary		ANOVA		Coefficients		
	R	R Square	F	Sig.	B	Beta	Sig.
H1.1	.468 ^a	.219	135.625	.000	.189	.468	.000
H1.2	.465 ^a	.217	133.822	.000	.185	.465	.000
H1.3	.482 ^a	.233	146.793	.000	.194	.482	.000
H1.4	.446 ^a	.199	119.940	.000	.179	.446	.000
H1.5	.462 ^a	.214	131.406	.000	.187	.462	.000
H1.6	.475 ^a	.225	140.869	.000	.192	.475	.000

Note. Linear regression analysis. Source: (Author, 2024).

Given this violation, the t-test was adjusted by using the Welch correction, which is robust to unequal variances. The corrected analysis continued to show a significant difference between the two groups,

reinforcing the conclusion that company type plays a significant role in sustainability adoption. This adjustment ensures the reliability of the results despite the assumption violation.

The t -test findings ($[t(284.472) = -2.856, p = 0.005]$, 95% C.I. = $[-0.312, -0.057]$) indicate that, under the assumption of unequal variances, there is a significant difference between the Local and International groups. Notably, the mean value of the international service providers was higher compared to the local, and the magnitude of differences in the means (M difference = -0.185 , 95% C.I: $[-0.312$ to $-0.057]$) was significant, suggesting that international companies are more significant than local ones, as displayed in Table 62. In addition, the effect size was measured using Cohen's d ($d = -0.288$, 95% C.I: $[-0.477$ to $-0.098]$) as both groups have same sample size with no significant difference in the standard deviation (SD), suggesting a small effect as displayed in Table 7 (Gignac et al., 2016).

Table 7—Group Statistics Based for Company Type.

Group Statistics					
Company Type		N	Mean	Std. Deviation	Std. Error Mean
Adoption of Sustainability	local	161	5.07	0.699	0.055
	International	325	5.26	0.612	0.034

Note. Source: (Author, 2024).

Table 8—Independent Sample t-test for Company Type and Adoption of Sustainability.

Independent Samples Test									
		Levene's Test for Equality of Variances				t-test for Equality of Means			
		F	Sig.	t	Df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	95 percent Confidence Interval of the Difference
									Lower Upper
Adoption of sustainability	Equal variances assumed	4.297	.039	-2.986	484	.003	-.185	.062	-.306 -.063
	Equal variances not assumed			-2.856	284.472	.005	-.185	.065	-.312 -.057

Note. Source: (Author, 2024).

Table 9—Effect Sizes Company Type and Adoption of Sustainability.

		Independent Samples Effect Sizes			
		Standardizer ^a	Point Estimate	95% Confidence Interval	
				Lower	Upper
Adoption of Sustainability	Cohen's d	0.642	-0.288	-0.477	-0.098
	Hedges' correction	0.643	-0.287	-0.477	-0.098
	Glass's delta	0.612	-0.302	-0.492	-0.111

Note. a. The denominator used in estimating the effect sizes.

Cohen's d uses the pooled standard deviation.

Hedges' correction uses the pooled standard deviation, plus a correction factor.

Glass's delta uses the sample standard deviation of the control group. Source: (Author, 2024).

Accordingly, based on the observed significant differences in sustainability adoption among oil and gas service provider companies in the UAE with respect to company type, the alternative hypothesis (H2_a)—which posits a significant relationship between company type and sustainability adoption—is supported, while the null hypothesis (H2_o) is rejected

Discussion

A demographic analysis of the 486 survey responses revealed notable patterns. The largest group of participants fell within the 31–40 age range, comprising 37.2% of respondents (181 individuals). Males represented the majority of the sample at 71.2% (346 respondents), while females accounted for 28.8% (140 respondents). Regarding work experience, the most represented group consisted of individuals with over 30 years of experience (35.2%, 171 respondents), reflecting a well-informed and seasoned workforce. Educationally, the majority held a bachelor's degree (71.2%, 346 respondents). Professionally, technical roles dominated the sample, making up 55.3% (269 respondents), indicating a strong operational focus among participants. Analysis by company type showed that approximately 70% of responses (325 out of 486) originated from international service providers

This study proposed and tested seven hypotheses to provide a comprehensive analysis of sustainability adoption among oil and gas (O&G) service providers in the United Arab Emirates (UAE). All null hypotheses were rejected in favor of their respective alternative hypotheses, indicating statistically significant relationships among the examined variables. The first six sub-hypotheses addressed the influence of various dimensions of relative advantage on sustainability adoption, thereby directly responding to the first research question. The seventh hypothesis assessed whether company type—local versus international—affects sustainability adoption, and the results confirmed a significant difference between the two, with international companies demonstrating higher adoption levels.

The linear regression analysis identified the most influential drivers of sustainability adoption among the six dimensions of relative advantage. Market uncertainty emerged as the leading factor, accounting for 48.2% of the variance in sustainability adoption. It was followed by market competition (47.5%), financial incentives (46.6%), market share (46.5%), and client perception (46.2%). Return on investment (ROI) was the lowest contributor, though still significant, at 44.6%. Finally, the company type emerged as a significant factor affecting the sustainability adoption where international companies were promoted. For the independent samples t-test, assumptions of normal distribution and homogeneity of variances were verified using the Shapiro-Wilk and Levene's tests, respectively. The results revealed a statistically

significant difference in sustainability adoption between local and international companies ($t = 3.27$, $p = 0.001$), suggesting that company type does play a meaningful role in shaping adoption outcomes. These insights highlight key areas where O&G service providers should concentrate their efforts to foster sustainability and align with the UAE's strategic goal of achieving Net Zero carbon emissions.

Conclusion, Recommendations, and Future work

While the term "sustainability" is widely used, there is a need for practical investigations that showcase the experiences, both positive and negative, of adopting sustainability to provide guidance and not only theoretical but practical results to end users in this field. This research aimed to fulfill that need and provided valuable insights for those considering engaging in sustainability practices. Overall, the research concluded that there are seven relative advantages factors that play a vital role in motivating oil and gas service providers in the UAE to adopt sustainability practices. The type of the company either national or international was also key identifier of the adoption process, international companies seem to be more active.

Consequently, this led to formulation of a roadmap which can be used to accelerate the progression wheel toward sustainability within an interrelated important business, oil and gas, that is merely considered the most complicated topic in the energy sector. The implications of this study are directed to 1) the facilitators; the entity which has the authority to enact laws and create changes with respect to sustainability implementation, or 2) the potential adopters; the entity that might find its empowering keys and prosperity by changing the way of doing business and transform to sustainability based on the running circumstances.

Specifically, this study was limited to oil and gas service provider companies which unfolded only one side of the complete business cycle. Although it was very helpful to highlight the drivers and inhibitors for this type of companies which are mostly considered facing the highest difficulties due to their work nature. Investigating the sustainability adoption with state owned companies in UAE will complement the other side and could bring to attention more hidden and worthy concerns. Looking from a wide angle and considering the fact that all gulf countries almost sharing the same work environment and business activities, it would be deserving step to confirm the UAE's sustainability adoption results with other neighborhood countries which open the door for other future regional sustainability cooperation.

Finally, the decision for sustainability shift might be free of choice now, however the expected challenges in the upcoming years with regards to environment, climate change, GHG, poverty, supply chain, food and medicine, global sea level, and others definitely going to apply high pressure on lagging entities, the switch is inevitably coming.

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Nomenclature

ADNOC	Abu Dhabi National Oil Company
AVE	Average Variance Extracted
(O&G)	Oil and Gas
CR	Composite Reliability
GHG	Greenhouse Gasses

NOAA National Oceanic and Atmospheric Administration
 RAD Relative Advantage
 ROCE Return on Capital Employed
 ROI Return on Investment
 UAE United Arab Emirates

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